Pharmaceutical Analytical Chemistry II
PCC/PCD 105

Week 4 lecture

Oxidation Reduction Reactions

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The Nernst Equation

- Consider the half-reaction:

$$aA + bB + ne^- \Leftrightarrow cC + dD$$

Reduced form

$$E = E^0 - \frac{0.0592}{n} \log \frac{[C]^c [D]^d \ldots}{[A]^a [B]^b \ldots} = \frac{[\text{Products}]}{[\text{Reactants}]}$$

Oxidized form

$$E = E^0 - \frac{0.0592}{n} \log \frac{[C]^c [D]^d \ldots}{[A]^a [B]^b \ldots}$$

Cell Potential Calculations

$$\boxed{E_{cell} = E_{cathode} - E_{anode}}$$

Nernst equation is used for measurement of:

1- Electrode potential at non standard conditions.

2- cell potential at non standardcconditions.

3- Keq

4- Titration curves

Factors affecting the value of the potential(E):

$$\uparrow E = E^0 - \frac{0.0592}{n} \log \frac{\downarrow [\text{products}] \ldots}{\uparrow [\text{reactants}] \ldots}$$

- Any factor that increases the products (numerator) will decrease E, and vice versa
- Any factor that increases the reactants (denominator) will increase E, and vice versa
- the factors include : pH, Complexation, precipitation and association.

Effect of pH

- <u>Strongly acidic solutions (pH ≤ 1)</u>

$$MnO_4^- + 8H^+ + 5e \rightarrow Mn^{2+} + 4H_2O$$

$\underline{E^0 = +1.51\ V}$

As hydrogen ions increase (lower pH value), E increases, used in selective oxidation.

$$\uparrow E = E^0 - \frac{0.0592}{5} \log \frac{[Mn^{2+}]}{[MnO_4^-]\,[H^+]^8 \uparrow}$$

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Effect of complexation

$$Fe^{3+} + e^{-} \leftrightarrow Fe^{2+}$$

$$\underline{E^0 = +0.77\ V}$$

As ferric ions decrease,E decreases, used in selective oxidation.

$$H_3PO_4 + Fe^{3+} \longrightarrow Fe\,(HPO_4)^{+}$$

$$\downarrow E = E^0 - \frac{0.0592}{1} \log \frac{[Fe^{2+}]}{[Fe^{3+}]\downarrow}$$

Effect of precipitation and association

- $Fe(CN)_6^{3-} + e^- \leftrightarrow Fe(CN)_6^{4-}$
- Precipitation with zinc or association in the prescence of HCL lead to lower conc. Of ferrocyanide, shift the reaction to the right, favoring reduction.

$$\uparrow E = E^0 - \frac{0.0592}{1} \log \frac{[\textbf{Ferrocyanide}]\downarrow}{[\textbf{Ferricyanide}]}$$

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In any reaction, reactants concentrations decrease by time and products concentration increase.

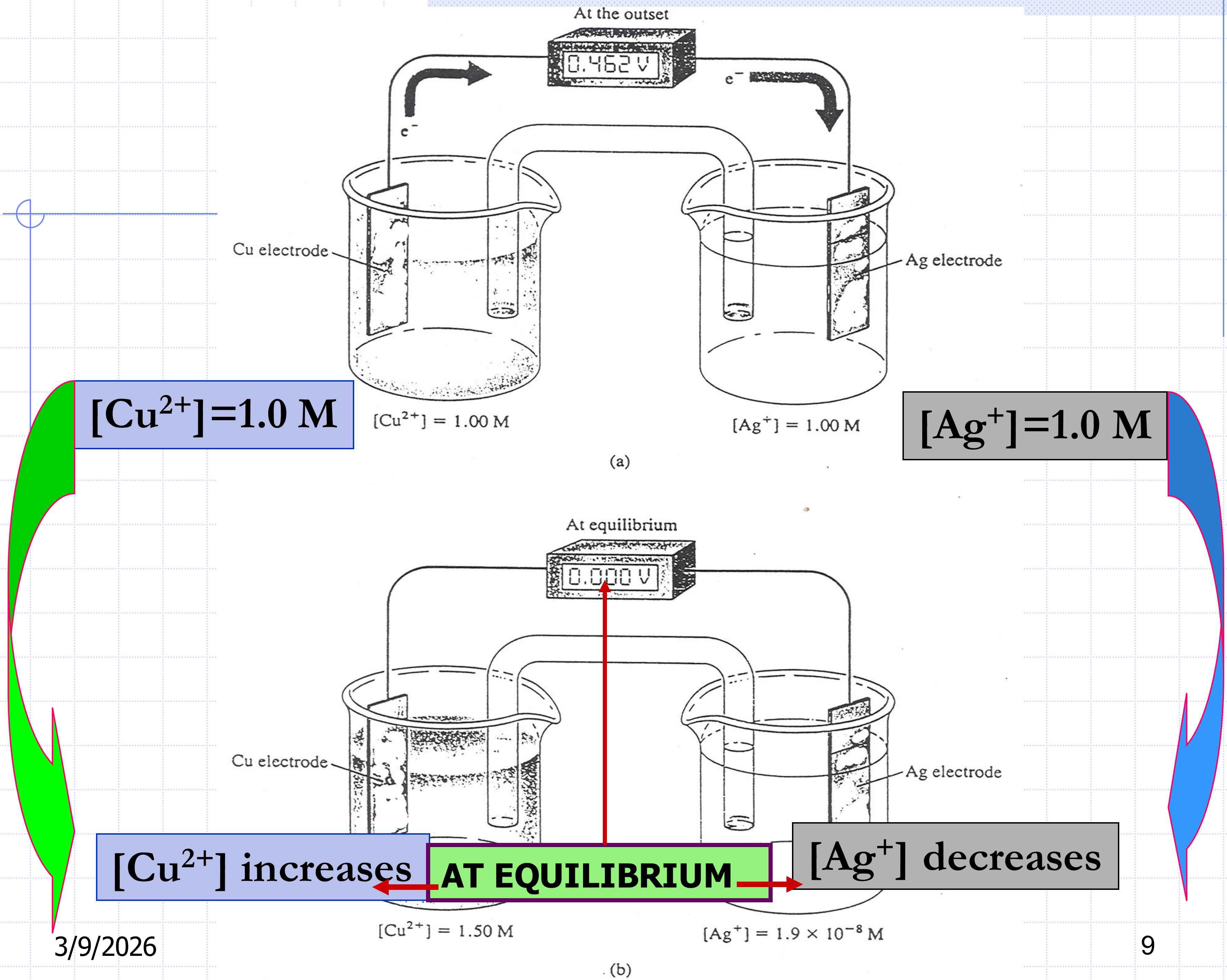
Start $\longleftrightarrow$ At equilibrium Keq

Q Reaction quotient $= \dfrac{\textit{Products}}{\textit{Reactants}}$:

Q= Keq ………………At equilibrium

Q< Keq ………………Forward reaction predominates

Q> Keq……………… Reverse reaction predominates


At the outset
0.462 V
e⁻
e⁻
Cu electrode
Ag electrode
[Cu^{2+}] = 1.00 M
[Ag^{+}] = 1.00 M
(a)
At equilibrium
0.000 V
Cu electrode
Ag electrode
[Cu^{2+}] = 1.50 M
[Ag^{+}] = 1.9 × 10^{-8} M
(b)
[Cu^{2+}]=1.0 M
[Ag^{+}]=1.0 M
[Cu^{2+}] increases
AT EQUILIBRIUM
[Ag^{+}] decreases

EQUILIBRIUM CONSTANTS FOR OXIDATION / REDUCTION REACTIONS

$$Cu(s) + 2Ag^{+} \Leftrightarrow Cu^{2+} + 2\,Ag(s)$$

$$K_{eq} = \frac{[Cu^{2+}]}{[Ag^{+}]^{2}}$$

Under equilibrium conditions, *the potential of the cell becomes zero. Thus, at chemical equilibrium,*

$$E_{cell} = 0.0 = E_{cathode} - E_{anode} = E_{Ag^{+}} - E_{Cu^{2+}}$$

$$E_{cathode} - E_{anode} = E_{Ag^{+}} - E_{Cu^{2+}} = 0.0$$

$$E_{cathode} = E_{anode}$$

Calculation of Equilibrium Constants

$$Cu(s) + 2Ag^{+} \Leftrightarrow Cu^{2+} + 2\,Ag(s)$$

$$\frac{2(E^{0}_{Ag^{+}} - E^{0}_{Cu^{2+}})}{0.0592} = \log \frac{[Cu^{2+}]}{[Ag^{+}]^{2}} = \log K_{eq} \qquad (6)$$

$$\frac{n(E^{0}_{\text{Cathode}} - E^{0}_{\text{Anode}})}{0.0592} = \log K_{eq}$$

Example 1

- Calculate the equilibrium constant for the reaction,

$$Cu(s) + 2Ag^{+} \Leftrightarrow Cu^{2+} + 2\,Ag(s)$$

$$\log K_{eq} = \frac{2\,(0.799 - 0.337)}{0.0592} = 15.6$$

$$K_{eq} = \text{antilog}\ 15.6 = 4.1 \times 10^{15} \approx 4 \times 10^{15}$$

OXIDATION/REDUCTION INDICATORS

1. General Redox Indicators

- **True oxidation/reduction indicators are substances that change color upon being oxidized or reduced.**
- The **color changes** of true redox indicators **depends upon the changes in the potential of the system** that occur as the titration progresses.
- The **half-reaction** responsible for color change in a typical general redox indicator can be written as,

$$In_{ox} + ne \Leftrightarrow In_{red}$$

Color A   **Color B**

$$E = E^0 - \frac{0.0592}{n} \log \frac{[In_{red}]}{[In_{ox}]}$$

- A change from the color of the **oxidized form** of the indicator to the color of the **reduced form** requires a **change of about 10 in the ratio of reactant concentrations**; that is, a color change is seen when,

$$\frac{[In_{red}]}{[In_{ox}]} \leq \frac{1}{10}$$

changes to,

$$\frac{[In_{red}]}{[In_{ox}]} \geq \frac{10}{1}$$

- The potential change required to produce the full color change of a typical general indicator can be found by substituting these two values into Equation 10, which gives,

$$E = E^{0}_{\mathrm{In}} \pm \frac{0.0592}{n}$$

- This equation shows that a typical general indicator exhibits a detectable color change when a titrant causes the system potential to shift from $E_{\mathrm{In}} + 0.0592/n$ to $E_{\mathrm{In}} - 0.0592/n$, or about $(0.118/n)$ V. For many indicators, ***n* = 2** and a change of **0.0592 V** is thus sufficient.

Table 2

Selected Oxidation/Reduction Indicators*

Indicator	Color		Transition	Conditions
	Oxidized	Reduced	Potential, V	
5-Nitro-1,10-phenanthroline iron(II) complex	Pale blue	Red-violet	+1.25	1 M H_2SO_4
2,3′-Diphenylamine dicarboxylic acid	Blue-violet	Colorless	+1.12	7–10 M H_2SO_4
1,10-Phenanthroline iron(II) complex	Pale blue	Red	+1.11	1 M H_2SO_4
5-Methyl 1,10-phenanthroline iron(II) complex	Pale blue	Red	+1.02	1 M H_2SO_4
Erioglaucin A	Blue-red	Yellow-green	+0.98	0.5 M H_2SO_4
Diphenylamine sulfonic acid	Red-violet	Colorless	+0.85	Dilute acid
Diphenylamine	Violet	Colorless	+0.76	Dilute acid
p-Ethoxychrysoidine	Yellow	Red	+0.76	Dilute acid
Methylene blue	Blue	Colorless	+0.53	1 M acid
Indigo tetrasulfonate	Blue	Colorless	+0.36	1 M acid
Phenosafranine	Red	Colorless	+0.28	1 M acid

*Data in part from I. M. Kolthoff and V. A. Stenger, *Volumetric Analysis*, 2nd ed., Vol. 1, p. 140 New York: Interscience, 1942.

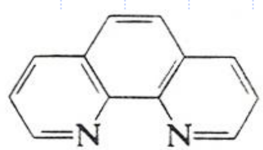
Iron(II) Complexes of Orthophenathrolines

- A class of organic compounds known as **1,10-phenanthrolines** (or **orthophenanthrolines**) form stable complexes with iron(II) and certain other ions.
- Three orthophenanthroline molecules combine with each iron ion to yield the complex "ferroin $(phen)_3Fe^{2+}$".
- The complexed iron in the ferroin undergoes a reversible oxidation/ reduction reaction that can be written as,

$$\mathbf{(Phen)_3Fe^{3+}} + e^- \Leftrightarrow \mathbf{(Phen)_3Fe^{2+}} \qquad E^0 = +1.06\ \mathrm{V}$$

Pale Blue — Red

Feriin — Ferroin

3  $+ Fe^{2+} \longrightarrow [(C_{12}H_8N_2)_3Fe]^{2+}$

This complex is readily oxidised reversibly to the corresponding *ortho*-phenanthroline–iron(III) complex, which is pale blue in colour:

$$[(C_{12}H_8N_2)_3Fe]^{2+} \rightleftharpoons [(C_{12}H_8N_2)_3Fe]^{3+} + e$$

Diphenylamine and its Derivatives

- Diphenylamine ($C_{12}H_{11}N_2$), in the presence of a strong oxidizing agent, is believed to undergo two reactions.
- The first reaction is irreversible to diphenylbenzidine

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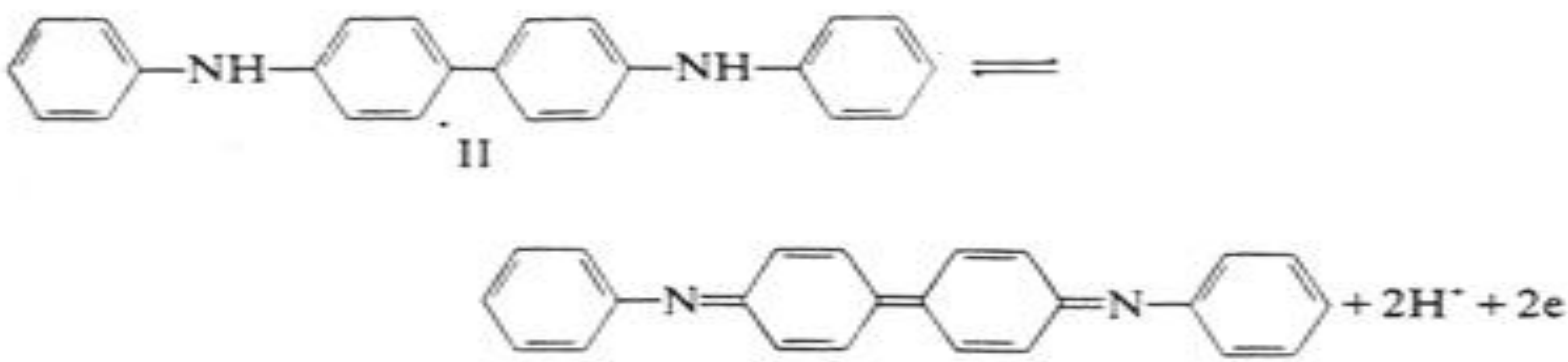
- But the second can be reversible and constituted the actual indicator reactions.
- The reduction potential for the second reaction is about +0.76 V.

Diphenylbenzidine (II) is colourless, but is reversibly oxidized into diphenylbenzidine violet (III):



II

III

*Diphenylamine is not very soluble in water, and indicator solutions must be prepared in rather concentrated sulfuric acid.

*The sulfonic acid derivative of diphenylamine, which has the same structure, does not suffer from these disadvantages.
The color change is somewhat sharper, however, passing from colorless through green to deep violet. The transition potential is about +0.8 V and again is independent of the acid concentration.